

Dhvani Dave

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Software Engineer who enjoys building meaningful products, solving problems and learning along the way.

EDUCATION

University at Buffalo

Master of Science in Computer Science; GPA: 3.9/4.0

Buffalo, NY

Aug. 2025 – Dec. 2026

Relevant Coursework: Algorithms, Machine Learning, Deep Learning, Operating Systems

University of Mumbai

Bachelor of Engineering in Computer Engineering; GPA: 3.6/4.0

Mumbai, India

Aug. 2019 – Jun. 2023

EXPERIENCE

Cloud Software Engineer

Jul. 2023 – Apr. 2025

CloudRock

Mumbai, India

- Delivered an enterprise **Workday** implementation for a major insurance client, owning the technical lifecycle from requirements and system design through integration, testing, and go-live.
- Built XML/XSLT data transformation pipelines in **Workday Studio** and custom EIB templates to automate scheduled batch data exchange between Workday and downstream HR/Finance systems.
- Implemented custom validation and business rule logic and wrote **Python** scripts to reconcile and pre-validate bulk data loads, reducing data errors by **30%** and cutting integration processing time.
- Designed and executed unit, integration, and regression test suites, and supported the post go-live hypercare phase by triaging and resolving production issues through **Jira** with cross-functional HR, Finance, and QA teams.

Software Engineer Intern

Aug. 2022 – May 2023

Vansh Consulting Inc.

Mumbai, India

- Developed an internal client-operations portal in **Python (Django)** and **React with TypeScript**, replacing spreadsheet-based tracking of supplier and purchase-order records for a team of ~15 consultants.
- Built **REST APIs** and scheduled sync jobs that pulled order and vendor data from client ERP and CRM systems into **PostgreSQL**, adding indexes and rewriting slow joins to cut the nightly run from ~**45** to ~**15 minutes**.
- Containerized the application with **Docker** and set up a **GitHub Actions** pipeline for build, test, and deploy, writing ~60 unit and integration tests and replacing a manual release process that had caused repeated environment issues.
- Prototyped a support-ticket triage service in Python using **spaCy** and **scikit-learn** that classified incoming requests into 6 categories at ~**85% accuracy**, reducing manual sorting for the helpdesk team.

PROJECTS

MiniDyn – Rigid Body Dynamics Simulator | C++, CMake, GoogleTest, WebAssembly, Next.js

Jul. – Aug. 2026

- Built an object-oriented 3D multibody dynamics engine in modern **C++** with a layered architecture: a stable public API over pluggable force elements and Baumgarte-stabilized joint constraints, so new physics models integrate without touching the solver core.
- Compiled the solver to **WebAssembly** via Emscripten/Embind to drive a real-time Next.js dashboard surfacing energy-conservation and constraint-error telemetry, and verified numerical consistency with a **GoogleTest** suite run through a **CMake** and **GitHub Actions** pipeline.

SentinelAI – Coral Hackathon | FastAPI, Next.js, GraphQL, PostgreSQL, Neo4j

Apr. – May 2026

- Architected a distributed 6-agent backend pipeline (**FastAPI**) integrating **PostgreSQL** and **Neo4j** for graph-based fraud detection, exposing a **GraphQL** API consumed by the Next.js frontend for investigation reporting.
- Validated the system end-to-end on a case with 8 high-risk transactions, producing a fraud risk score of 100 with accurate relationship detection across the transaction graph.

CodeSense – Agentic AI Developer Copilot | FastAPI, React, LangGraph, FAISS, Tree-sitter, Qwen3-8B

Jan. – Mar. 2026

- Designed, coded, and deployed a full-stack agentic debugging assistant (**FastAPI** backend, **React** frontend) that retrieves code context, detects bugs, and generates validated patches across the full development lifecycle.
- Experimented with large language models (LLMs) by fine-tuning **Qwen3-8B** with **QDoRA** on 5,400 samples, improving **SWE-bench Token F1 from 0.075 to 0.257** and reducing hallucinations by **93%**.

TECHNICAL SKILLS

Languages: Python, Go, Java, C, C++, TypeScript, JavaScript, SQL, CMake, WebAssembly

AI/ML: PyTorch, TensorFlow, Hugging Face, scikit-learn, spaCy, NumPy, Pandas, LangGraph, RAG, FAISS, Fine-Tuning (QDoRA)

Backend: FastAPI, Django, Node.js, REST, GraphQL, Microservices | **Frontend:** React.js, Next.js, Streamlit

Data & Infra: PostgreSQL, MongoDB, Neo4j, AWS (EC2, S3, Lambda), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux

PROFESSIONAL CERTIFICATIONS

Building with the Claude API (*Anthropic*) | Introduction to Model Context Protocol (MCP) (*Anthropic*) | Build RAG Applications: Get Started (*IBM*) | BCG GenAI Job Simulation (*Forage*)